Math 54 — Week 12

Solutions to the problem set

P1. Idea. $p(r) = r^2 - r - 2 = (r-2)(r+1)$, so 0 is not a root and $s = 0$: guess a polynomial of degree 1 with both terms.

1. With $y = At + B$: $y' = A$, $y'' = 0$, so
$$y'' - y' - 2y = -A - 2At - 2B = -2At + (-A - 2B).$$
2. Matching $4t$: $-2A = 4$ and $-A - 2B = 0$, so $A = -2$, $B = 1$, and $y_p = -2t + 1$.
3. Check: $0 - (-2) - 2(-2t + 1) = 4t$.

P2. Idea. i is not a root of $p(r) = r^2 + 2r + 5$ (roots $-1 \pm 2i$), so $s = 0$; the complex method replaces a 2×2 system by one division.

1. **Real coefficients.** With $y = A\cos t + B\sin t$: $y' = -A\sin t + B\cos t$, $y'' = -A\cos t - B\sin t$, and
$$y'' + 2y' + 5y = (4A + 2B)\cos t + (4B - 2A)\sin t.$$
Matching $10\cos t$: $4A + 2B = 10$, $-2A + 4B = 0$, so $A = 2B$, $10B = 10$: $B = 1$, $A = 2$, and $y_p = 2\cos t + \sin t$.
2. **Complex exponential.** $p(i) = -1 + 2i + 5 = 4 + 2i$, so $z = \big(10/(4+2i)\big)e^{it}$ solves $L[z] = 10e^{it}$, and $y_p = \operatorname{Re} z$:
$$\frac{10}{4+2i} = \frac{10(4-2i)}{20} = 2 - i, \qquad z = (2-i)(\cos t + i\sin t), \qquad \operatorname{Re} z = 2\cos t + \sin t.$$

P3. Idea. $p(r) = r^2 - 4$ has the simple root 2, the exponent on the right, so $s = 1$; the shortcut "exponent is a simple root" gives the answer at once.

1. $p(2) = 0$ and $p'(2) = 2 \cdot 2 = 4 \neq 0$. By the key formula with $q = t$, $L[te^{2t}] = p'(2)e^{2t} = 4e^{2t}$.
2. So $y_p = \frac{8}{4}te^{2t} = 2te^{2t}$.
3. Check: $y_p' = (2 + 4t)e^{2t}$, $y_p'' = (8 + 8t)e^{2t}$, and $y_p'' - 4y_p = 8e^{2t}$.

P4. Idea. $p(r) = (r+3)^2$ has the double root -3, so $s = 2$ and $L[qe^{-3t}] = q''e^{-3t}$.

1. By the double root shortcut ($L[qe^{rt}] = aq''e^{rt}$ at a double root), we need $q'' = 2$, so $q = t^2$ and $y_p = t^2e^{-3t}$.
2. Check: $y_p' = (2t - 3t^2)e^{-3t}$, $y_p'' = (2 - 12t + 9t^2)e^{-3t}$, and
$$y_p'' + 6y_p' + 9y_p = \big(2 - 12t + 9t^2 + 12t - 18t^2 + 9t^2\big)e^{-3t} = 2e^{-3t}.$$

P5. Idea. $p(r) = r^2 + 4$ has roots $\pm 2i$; compare each exponent ($2i$, $2 + i$, 2) with these roots.

1. (a) $t\sin 2t$: $\alpha + i\beta = 2i$ is a root, so $s = 1$, and $m = 1$:
$$y_p = t(A_1t + A_0)\cos 2t + t(B_1t + B_0)\sin 2t.$$
2. (b) $e^{2t}\cos t$: $2 + i$ is not a root, $s = 0$: $y_p = Ae^{2t}\cos t + Be^{2t}\sin t$.
3. (c) $3t^2e^{2t}$: 2 is not a root, $s = 0$; $\cos 2t$: $2i$ is a root, $s = 1$. By superposition,
$$y_p = (A_2t^2 + A_1t + A_0)e^{2t} + t(C\cos 2t + D\sin 2t).$$

P6. **Idea.** $6t - 4e^t = 3(2t) - 4(e^t)$, so superposition gives a particular solution, and the homogeneous solutions supply the rest.

1. By superposition, this combination solves the equation:
$$3y_1 - 4y_2 = 3\Big(-t + \frac{1}{2}\Big) - 4\Big(-\frac{1}{2}e^t\Big) = -3t + \frac{3}{2} + 2e^t.$$
2. $p(r) = r^2 - r - 2 = (r-2)(r+1)$, so the homogeneous solutions are $c_1e^{2t} + c_2e^{-t}$. By Nagle Thm. 4, all solutions are
$$y = -3t + \frac{3}{2} + 2e^t + c_1e^{2t} + c_2e^{-t}.$$

P7. **Idea.** Find y_p (a polynomial, since 0 is not a root of $r^2 - 1$), add the homogeneous solution, then apply the conditions to the whole sum.

1. $y = At + B$ gives $y'' - y = -At - B = 2t$, so $A = -2$, $B = 0$: $y_p = -2t$.
2. The roots of $r^2 - 1$ are ± 1, so $y = -2t + c_1e^t + c_2e^{-t}$.
3. $y(0) = c_1 + c_2 = 0$ and $y'(0) = -2 + c_1 - c_2 = 1$ give $c_1 = \frac{3}{2}$, $c_2 = -\frac{3}{2}$:
$$y = -2t + \tfrac{3}{2}(e^t - e^{-t}) = -2t + 3\sinh t.$$

P8. **Idea.** $2i$ is not a root of $r^2 + 1$, so $y_p = A\cos 2t + B\sin 2t$; the homogeneous part oscillates at frequency 1.

1. For $y_p = A\cos 2t + B\sin 2t$, $y_p'' + y_p = -3A\cos 2t - 3B\sin 2t$, so $A = -\frac{1}{3}$, $B = 0$.
2. $y = -\frac{1}{3}\cos 2t + c_1\cos t + c_2\sin t$; $y(0) = -\frac{1}{3} + c_1 = 0$ and $y'(0) = c_2 = 0$ give
$$y = \tfrac{1}{3}(\cos t - \cos 2t).$$
3. The motion is a sum of two oscillations with frequencies 1 and 2; it is periodic with period 2π and stays bounded ($|y| \le \frac{2}{3}$), with no damping and no resonance (the forcing frequency 2 differs from the frequency 1 of the homogeneous solutions $\cos t$, $\sin t$).